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Targeting climate finance for global forests

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Abstract

Comprehensive data on the costs of climate change mitigation are needed to direct the scale and distribution of climate finance to the sectors and regions where it will be most cost effective. We estimate finance needed to achieve regional forest-based mitigation targets aggregated from Nationally Determined Contributions (NDCs) to the Paris Agreement, submitted between 2015 - 2021. We compare the costs of achieving these regional targets to a Global Coordination scenario in which a similar level of finance is available as is required to meet upper bound NDC targets, but which allows mitigation to occur in regions where it is lowest cost. We find that regions accounting for 70% of global forest carbon stocks can meet or exceed their forest-based NDC targets with carbon prices less than $100 / ton CO_2. Total investment required to meet these goals is $28-70 billion annually by 2030. Under a Global Coordination scenario, we project nearly 3x the quantity of mitigation as in the upper bound NDC scenario, under similar levels of investment, in 2030. These findings demonstrate an opportunity for increasing mitigation action among regions with low-cost abatement potential that is not reflected in current NDCs, as well as substantial cost savings via policy solutions that target low-cost mitigation opportunities. This assessment provides insight into the magnitude and distribution of forest climate finance needed to reach current mitigation targets and informs the development of the next generation of NDCs.

Introduction

Avoiding forest conversion, restoring degraded and deforested landscapes, and improving forest management are critical to limiting global climate change. These and other land-based mitigation activities have the potential to achieve up to one-third of the cost-effective mitigation needed to limit warming to 1.5°C (*1*, *2*).

Abatement in forest ecosystems is also a prominent component of climate change mitigation strategies proposed by individual countries (*3*, *4*). More than 100 countries representing at least 90% of global forest cover explicitly include mitigation from forests in their nationally determined contributions (NDCs) (*5*).

Estimates of the potential mitigation from these forest-based components of NDCs range from 0.9 - 3.8 Gt CO_2 in 2030 (*3*, *4*, *6*). Importantly, these estimates do not account for market dynamics such as competition for land or forest product price effects, which could greatly influence total aggregate costs of achieving these targets, and therefore their likelihood of attainment (*7*). Models that represent these market and land dynamics are essential for estimating the scale of economically feasible mitigation opportunities (*8*).

The current scale of climate finance for forests, estimated at $2.3 billion annually (*9*), is at least an order of magnitude smaller than what is needed to achieve mitigation compatible with a 1.5°C pathway (*10*, *11*). It is evident that climate finance for forests needs to increase, and a science-based understanding of the scale and distribution of the finance required to meet NDC goals is essential.

This study uses an economic model of global forests, the Global Timber Model (GTM) (*12*, *13*), to assess the finance needed to achieve forest-based NDC goals across 16 global regions. We project regional forest carbon fluxes in 2030 using GTM across a set of climate finance scenarios, considering future market conditions and biophysical characteristics of regional forests. We use these projections to estimate the cost of meeting regional forest-based NDC targets submitted to the UNFCCC between 2015 - 2021 (*4*, *14*).

We compare the cost of achieving regional forest-based NDCs to a Global Coordination scenario in which a similar level of climate finance is available, but without requiring regional NDC targets to be met. This allows us to assess the efficiency of the NDC targets relative to a theoretically optimal allocation of mitigation action and indicates opportunities for cost-effective increases in regional mitigation targets.

These findings provide critical input for the development of an effective New Collective Quantified Goal (NCQG) for climate finance, which will replace the $100 billion annual cross-sectoral climate finance target set in 2009. In addition, this study can inform the development of the next generation of NDCs, which will both strengthen 2030 commitments and outline new 2035 mitigation targets.

Results

The GTM projects that global forests will act as a net carbon sink of 1.1 $GtCO_2$/yr in 2030 in the absence of climate finance. If countries achieve their NDCs targets, this sink will increase to 1.5 - 2.8 Gt CO_2/yr in 2030, resulting in mitigation of 0.4 – 1.6 Gt CO_2 relative to the reference case.

We estimate that the US, Canada, China, Oceania, and Russia can achieve their forest-based NDC targets without the need for additional climate finance (Group A in Figure 1). The forests in each of these regions were a net sink of CO_2 in 2020 and GTM projects that the strength of this sink will increase through 2030 even in the absence of climate finance. This trend is due to projected changes in land use and management, including future increases in global demand for timber. In response to these demand signals, Russia and Oceania are projected to intensify forest management, China to continue expanding forest area, and the US to pursue both strategies.

In 2020 the forests in South America not including Brazil were a net source of emissions, and emissions from forests will continue to increase under the regions' aggregated NDC target; therefore, meeting this target does not require additional climate finance. Likewise, Sub-Saharan Africa's lower bound NDC commitment is a continuation of net emissions from forests, which GTM projects will continue without climate finance.

On the other hand, low levels of investment are needed to meet forest-based NDC targets in Central America, Brazil, Southeast Asia, South Asia, and Europe (in line with a carbon price of \$20/tCO_2), as well as Sub-Saharan Africa's upper bound NDC target (in line with a carbon price

of $5/tCO$_2$) (Group B in Figure 1). In the absence of climate finance incentives, GTM projects the forests in these regions will either become a larger net source or a smaller net sink by 2030. Large opportunities for avoided deforestation in the tropics, and improved forest management in Europe, result in low-cost attainment of regional forest-based NDCs.

Medium levels of investment are needed to reach forest-based NDC targets in East Asia and Japan (Group C in Figure 1). Forests in Japan are projected to continue net sequestration even without direct investments, though carbon prices $50/ tCO$_2$ are needed to increase sequestration sufficiently to meet its NDC target. These relatively higher mitigation prices in Japan are due to its limited capacity to expand forest area. Forests in East Asia are projected to become a net source of emissions by 2030, but with a carbon price of 35$/tCO$_2$ forests can become a net sink and the region will meet its NDC target.

GTM projects that the region combining northern and eastern Africa and the Middle East will maintain a small net sink in 2030 (Group D in Figure 1). Reaching the regions' lower bound NDC target, which includes a large mitigation commitment from Ethiopia, will require a carbon price of $150/tCO$_2$, and reaching the region's upper bound target will require a carbon price >$200/tCO$_2$.


(A)
OCEANIA
RUSSIA
RSAM
CHINA
CANADA
US
NDC Range
No Finance
Low Finance
Medium Finance
High Finance
(B)
CENT AMER.
SOUTH ASIA
EEU
WEU
BRAZIL
SE ASIA
SSAF
(C)
E ASIA
JAPAN
(D)
AFME
-3000
-2500
-2000
-1500
-1000
-500
0
500
1000
Annual Flux (MtCO2/yr)

123

Fig. 1 Regional net forest CO_2 flux in 2030 based on reported NDCs (green, range represents the lower and upper bounds if applicable to that region), compared to the projected net CO_2 flux under in the GTM baseline (no finance scenario, blue) and under low (<\$20/tCO_2, yellow) medium (\$35/tCO_2 – 75/tCO_2, orange), and high (>\$100/tCO_2, red) climate finance ranges. Regions are grouped by the estimated level of investment needed to achieve their aggregate forest-based NDC target, with regions in group A achieving their target without additional price incentives, group B achieving their target with low price incentives, group C achieving their targets with medium price incentives, and group D achieving their targets with high prices incentives. *Notes:* In the figure RSAM refers to South America not including Brazil, SSAF to Sub-Saharan Africa, EEU to Eastern Europe, WEU to Western Europe, and AFME to North Africa and the Middle East. Positive values indicate net emissions from forests while negative values represent net sequestration from forest.

Globally we estimate that achieving all forest-based NDC commitments will require between \$28 – 70 billion annually in 2030. This assumes that the region comprised of northern and eastern Africa and the Middle east could meet its upper bound NDC target at a carbon price of 200\$/tCO_2. Southeast Asia, Brazil, and Northern Africa and the Middle East together comprise 67 - 85% of total finance needed to reach all NDC targets (Figure 2). On the other hand, due to the small differences between regional NDC targets and baseline forest carbon flux projections in Canada, Russia, China, South America outside Brazil, Oceania, and the US, these regions require no climate finance to reach their NDC targets.

We compare the costs of achieving regional NDC targets to a Global Coordination scenario, in which a similar level of total global climate finance is made available as is required to meet the upper bound NDC target, but without requirements as to where mitigation occurs. This is

simulated using a carbon price of \$20/tCO_2 that results in total finance of \$76 Billion annually in 2030. Under this Global Coordination scenario, total mitigation from forests reaches 4.6 GtCO_2, or almost three times more than the quantity of mitigation expected under achievement of upper bound NDC targets (1.6 Gt CO_2). Yet, finance needs under the Global Coordination scenario are \$76 Billion annually in 2030, just 8% greater than we estimate is needed to meet upper bound NDC targets in 2030 (\$70 Billion per year).

These cost efficiencies are the product of a geographic shift in mitigation under the Global Coordination scenario, with large increases in low-cost mitigation in some regions, and slight decreases in very high-cost mitigation in other regions, relative to regional NDC mitigation targets (Figure 2). Under the \$20 / tCO_2 Global Coordination scenario, we project that Canada, Russia, China, South America outside Brazil, Oceania, and the US begin to take advantage of low-cost abatement opportunities that are not reflected in the in the first generation of NDCs. Together, these six regions are projected to reach 1.6 Gt CO_2 of mitigation in 2030, requiring finance of \$17 Billion. We also project substantially more mitigation in Sub Saharan Africa under the Global Coordination scenario, reaching 1.1 Gt CO_2 with climate finance of \$19 Billion annually in 2030, and in Brazil, reaching 1.1 GT CO_2 with climate finance of \$17 Billion annually. On the other hand, we project slightly less mitigation in Japan, East Asia, and Northern Africa and the Middle East, relative to these regions' aggregate NDC commitments, due to the comparatively higher cost of mitigation in these locations.

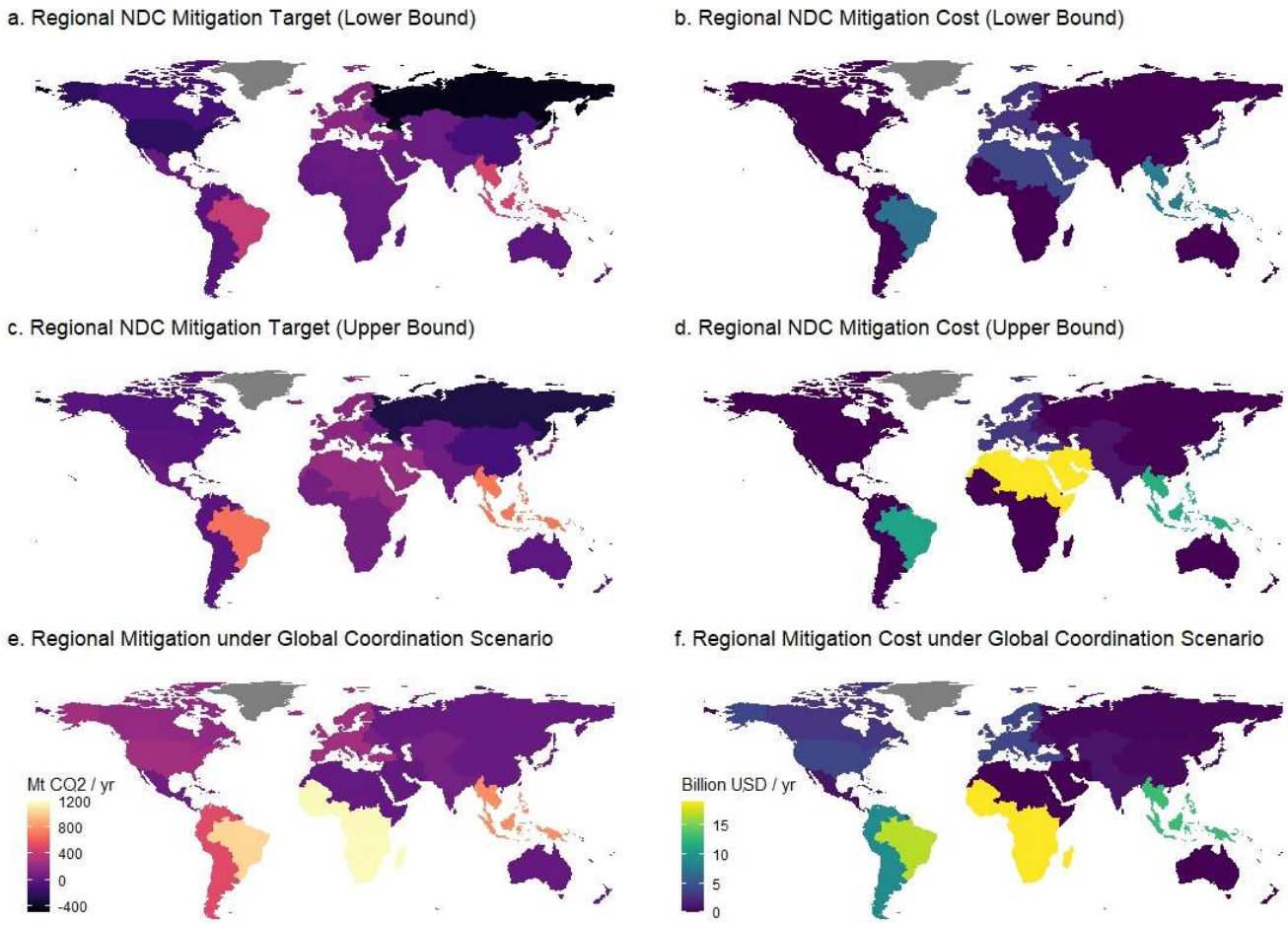


Figure 2. Maps of a. Regional forest-based mitigation under aggregate lower bound NDC targets in 2030, b. Regional cost of reaching aggregate lower bound NDC targets in 2030, c. Regional forest-based mitigation under aggregate upper bound NDC target in 2030, d. Regional cost of reaching aggregate upper bound NDC targets in 2030, e. Regional forest-based mitigation under a Global Coordination scenario simulated using a harmonized global carbon price of \$20/tCO_2, and d. Regional cost of reaching the Global Coordination scenario mitigation quantity.

Discussion

Our estimates of climate finance needed to realize forest-based NDC mitigation targets in 2030, of $28 - $70 billion annually, fall within the range reported elsewhere. These include estimates of $44 – $210 billion annually needed for terrestrial protection and restoration by 2030 (*15*), $45 – $65 billion needed for protection and restoration of nature in 2030 (*16*), and $19 – $32 billion annually to transition forests currently management for productive purposes to sustainable forestry practices (*17*). The UNFCCC Standing Committee on Finance estimated cross-sectoral climate finance needs of around $580 billion annually through 2030 to implement NDCs in developing countries (*18*). Our projections of mitigation costs for forest based NDCs in the tropics is roughly 10% of this estimate - a reasonable proportion given our narrow focus on forest-based mitigation, and the relative low-cost of forest-based abatement relative to other sectors of the economy.

This study makes two important advancements to inform climate finance for forests. First, we refine global scale analyses by estimating the distribution of finance needs needed to achieve regional forest-sector NDC targets. Overall, regions accounting for 70% of global forest carbon stocks can meet or exceed their forest-based NDC targets with carbon prices less than $100 / ton CO2. Yet on a regional basis we observe large differences in the costs of meeting NDC targets. Across the tropics and Europe, NDC commitments can be achieved with low levels of climate finance between $5 – 20/t$CO_2$. Smaller regions, including East Asia, Japan, and Northern Africa and the Middle East, have made commitments that will require medium- to high- levels of finance. While several key regions, including Canada, the US, Oceania, China, and South

America outside Brazil, have the capacity to meet their forest-based NDC targets without additional climate finance. Regions with additional low-cost mitigation potential beyond what is reflected in current NDCs are priorities for closing the gap between what has been pledged under NDCs and what is needed to reach emissions compatible with a 1.5°C pathway (*19*). High income countries in this group can lead by example in pledging more stringent emissions reductions from the forest sector and investing in additional abatement in the sector.

Next, this study identifies potential cost efficiencies in global forest-based mitigation, by comparing mitigation achieved by meeting the upper bound NDC target (1.6 Gt CO_2) to mitigation achieved under a Global Coordination $20 / tCO2 scenario (4.6 Gt CO_2). We project that this three-fold difference in mitigation can be achieved at costs of just 8% higher than the NDC scenario. This is due to less mitigation in regions with high-cost abatement (e.g., East Asia, Japan and Northern Africa and the Middle East), and substantially more mitigation in regions with low-cost abatement (e.g., Sub Saharan Africa, Brazil, South America outside of Brazil, Russia, Canada, and the US), under the Global Coordination scenario. This is consistent with previous studies showing that a theoretical global carbon market could result in mitigation nearly double current cross-sectoral NDCs without increasing total costs, compared to a scenario without international markets (*20*), as well as previous research demonstrating theoretical cost-savings via international mitigation transfers (*21–23*).

Our assessment is constrained by lack of clarity in forest-sector mitigation targets derived from submitted NDCs. These NDCs are widely acknowledged to be challenging to interpret given the broad diversity in how targets are articulated, the paucity of detail needed for clear

interpretation, and large uncertainties with respect to the estimation and reporting of LULUCF fluxes generally (*4*, *24*, *25*). Future iterations of NDC submissions are likely to have substantially improved clarity, facilitating improvements in these data inputs (*26*).

In addition, the estimates of mitigation costs presented in this study are based on a model which simplifies reality in several ways. For instance, theoretical costs of mitigation are estimated under the assumption that a uniform price incentive on carbon is applied to all forest-based activities, and land managers acted in the interest of optimizing their current and future welfare outcomes. These costs do not reflect transaction costs associated with developing and governing a high-integrity carbon market, building knowledge and capacity for participating in a market, or developing systems for monitoring, reporting and verifying carbon fluxes, which could be substantial (*27*). In practice international cooperation may be carried out on a piecemeal basis, with domestic regulations on allowable credit purchases, or bilateral arrangements that preference mitigation from certain geographies. Furthermore, there are non-monetary barriers to the adoption of mitigation activities in the land sector that are not accounted for in GTM and which can slow rates of adoption and behavior change needed to reach mitigation targets (*28*). These include for example cultural preferences for established land use and production patterns, uncertainty around new technologies or practices, or perceived volatility in carbon prices. These limitations are not unique to the GTM and given the central role that this and other economic models play in the assessment of climate mitigation scenarios and design of mitigation solutions, their continuous improvement merits attention and investment (*29*).

Despite these limitations, this study offers new insights into the magnitude and distribution of finance needed to reach current forest-based NDC targets, as well as opportunities for efficient allocation of abatement to maximize mitigation per dollar invested via global coordination. As current cross-sectoral NDC pledges are likely to fall short of limiting warming to 1.5°C (UNFCCC 2023), regional opportunities to cost-effectively increase forest-based mitigation, such as those highlighted here, are urgently needed.

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Online Methods

Regional NDC targets

Nationally Determined Contributions (NDCs) are central components of the United Nations Framework Convention on Climate Change (UNFCCC). They are voluntary mitigation pledges made by individual countries which outline specific climate action towards mitigating climate change and adapting to its impacts, reflecting each country's circumstances, capabilities, and development priorities.

Countries representing at least 90% of global forest cover included land use, land cover change, and forestry (LULUCF) in their NDC submissions (*5*, *30*). Countries formulated and reported their NDC targets using diverse approaches including variation in the articulation of targets (e.g., absolute quantities or change in emissions intensities), source and sink categories (e.g., forest definitions, activity definitions), reference points (e.g., relative to a historical year or relative to a projected baseline), and conditionality (e.g., whether financed domestically or requiring external support) (*3*, *24*, *25*). Of the 121 countries with explicit LULUCF mitigation targets articulated in the first round of NDC submissions, just 11 provided quantitative detail such as a specific LULUCF reduction target, or specific LULUCF mitigation measures (*5*).

This study uses estimates of LULUCF NDC targets for 44 countries (*4*, *14*), representing 84% of global forest cover (*31*) (**Table S1**). These studies of LULUCF NDC targets examined the text of NDCs submitted to the UNFCCC and extracted LULUCF specific targets in 2030 where available. Where needed they carried out additional analyses and modelling to determine the LULUCF-

specific contribution to a cross-sectoral NDC target. In some cases, NDC targets are presented as a range, reflecting uncertainty in the interpretation of the NDC and/or the difference between targets that are conditional or unconditional on international finance.

Most NDCs are not reported with sufficient detail to disaggregate mitigation targets from forests, wetlands, peatlands, grasslands, or from soil carbon sequestration on agricultural lands (*24*, *25*). We assumed that the mitigation needed to reach NDC targets is derived from forest-based activities including avoided deforestation, afforestation/ reforestation, and improved forest management. These activities comprise more than four-fifths of global land management mitigation potential, not including bioenergy with carbon capture and storage (*1*). Low-cost mitigation opportunities from other ecosystems, such as peatlands and wetlands, are not represented in this study. Excluding these mitigation opportunities from our assessment may lead to over-estimate of mitigation costs, particularly in regions such as Southeast Asia where peatlands play an important role in the regional mitigation portfolio (*32*). As NDC reporting transparency improves, future research will be able to differentiate mitigation targets, and improve estimates of mitigation costs, by ecosystem type (*33*).

Because GTM uses baseline greenhouse gas flux estimates calibrated to FAO reports (*31*), which differ from greenhouse gas inventory (GHGI) reported fluxes (*34*), we took steps to harmonize between the GHGI and GTM 2020 flux estimates. First, we calculated the difference between 2020 GHGI reported LULUCF fluxes and the 2030 country-reported NDC flux target (including upper and lower bounds, if provided as a range) for each country as follows:

$$Flux_{GHGI,2020} - Flux_{NDC,2030} = \Delta_{\boldsymbol{NDC,2030}}$$

We used the resulting difference to represent the change required in GTM between 2020 and 2030 to achieve the NDC target, rather than requiring GTM to reach a specific flux value that may be misaligned with the FAO calibration data. Specifically, we calculated the 'target flux' as follows:

$$Flux_{GTM,2020} - \Delta_{\boldsymbol{NDC,2030}} = TargetFlux_{GTM,2030}$$

We then summed country level target fluxes within each of the sixteen regions represented in GTM, to report regional targets harmonized to GTM initial flux values (**Table S2**). For countries that did not provide a LULUCF sector-specific target in their NDC, we assumed LULUCF emissions remain constant, neither contributing to nor detracting from the regional aggregate NDC target. This assumption may result in underestimation of mitigation costs, as countries that did not provide a LULUCF target may still have low-cost forest-based mitigation opportunities represented in GTM that can be 'contributed' to the regional mitigation target. However, given that the countries included in our assessment comprise 84% of global forest cover, the scale of underestimation due to regional aggregation is likely small.

The Global Timber Model

This study uses an economic model of the global forest sector, the Global Timber Model (GTM), to estimate mitigation quantities and costs under a reference scenario and under alternative scenarios representing varying levels of investments in forest-based mitigation activities. GTM has been used to project potential mitigation quantities and costs in several previous studies

(*11*, *35*, *36*), and the model's data, parameters, carbon accounting procedures, and structural equations have been documented (*37–39*).

GTM is a dynamic intertemporal economic optimization model that combines spatially detailed data on forests with an economic model that weighs forest management alternatives. The model optimizes the net present value of consumers' and producers' surplus in the forest sector, to project forest resource use projected macroeconomic conditions and policy futures. Forest sector abatement activities in one region may alter the quantity of forest products entering the global market, potentially leading – via changes in forest product prices - to changes in forest management and CO_2 fluxes in other regions. It is therefore critical to investigate potential mitigation and costs using a tool such as GTM that accounts for such market feedbacks and interactions, to develop a comprehensive picture of forest-based mitigation costs.

This study used a version of GTM including updates described in Favero et al. (*12*, *13*). This includes updates to the initial stock of carbon for the US and Europe using more recent data for these regions, as well as updated land elasticity and agriculture productivity parameters, which are likely to affect the costs of converting land to forestland. This study uses a version of GTM that includes carbon fertilization effects consistent with RCP 4.5 in the function that represents regional forest natural productivity (*40*, *41*).

The model does not include other potential impacts of climate change, which have the potential to substantially influence forest carbon flux dynamics and diminish the capacity of forests to contribute to mitigation (*42*). Incorporating estimates of these projections will be

critical for comprehensive assessment of the magnitude and costs of forest-sector mitigation, particularly along time frames beyond mid-century.

Scenarios

We used GTM to establish a reference level of future CO_2 fluxes from forests and forest products in the absence of incentives for forest-based mitigation. This reference scenario is calibrated to macroeconomic projections of GDP and population worldwide from the Shared Socioeconomic Pathways (SSP) scenario 2 and the 2020 U.S. Annual Energy Outlook.

CO_2 mitigation incentives are represented in the model in the form of rental payments for carbon sequestration and annual subsidies for carbon in timber products, under 8 scenarios with carbon payments starting at $5, $20, $35, $50, $75, $100, $150, and $200 and growing at 3% per year. We group these scenarios into low (<\$20/tCO_2) medium (\$35/tCO_2 – 75/tCO_2), and high (>\$100/tCO_2) climate finance ranges for the purpose of visualization and reporting.

There are no limits or constraints on forest-based mitigation activities at the country level outside biophysical limits already included in the model (e.g. trees could not be planted in areas that are not suitable under current climate conditions). For each scenario, the model estimates the optimal level of four potential mitigation activities in each region in response to carbon payments: avoided deforestation, forest management activities, increasing harvest rotations, and re/afforestation (including natural forest regeneration and intensively managed timber plantation establishment).

In the model investments in the form of carbon payments (*Cprice)* apply to CO2 emissions from forests. Specifically, the global mitigation volume (*A*) at any time (*t*) is measured as the difference between CO2 emissions in the baseline scenario without payments (E_{base}) and emissions estimated in the payment scenario (E_{cprice}) across regions (*i*).

$$A_t = \sum_i \left(E_{base,t,i} - E_{cprice,t,i}\right)$$

Investments (I_t) are calculated *ex-post* by discounting annual expenditures to compensate landowners for their forest-based mitigation activities as follow:

$$I_t = \sum_t \frac{A_t \times Cprice_t}{(1+r)^{t-t_0}}$$

where *r* is the discount rate of 5% and *t0* is the initial year. A similar approach was used in Austin et al. (*11*) and Favero et al. (*13*).

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Supplemental Information for "Targeting climate finance for global forests"

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Table S1: Nationally Determined Contribution LULUCF net greenhouse gas flux target in 2030 (Mt CO2/ yr). Based on analysis presented in Forsell et al. 2016 and den Elzen et al. 2022 (negative values indicate net sequestration, positive values indicate net emissions).

Country	2030 Nationally Determined Contribution net LULUCF emissions target (Mt CO2/yr)
China	-1099
United States	-1062 to -850
Russia	-524 to -377
India	-389 to -367
EU28	-310
Malaysia	-229
Central African Republic	-226 to -203
Mexico	-184
Mali	-153
Gabon	-107
Turkey	-106 to -104
Japan	-93
Vietnam	-71 to -59
South Korea	-65 to -46
Australia	-43
Bolivia	-39
Belarus	-37
Papua New Guinea	-32
Congo	-48 to -30

Mongolia	-26
New Zealand	-23
South Africa	-23
Ukraine	-23
Norway	-21
Zambia	-30 to -17
Senegal	-23 to -9
Saudi Arabia	-9
Chile	-7 to -8
Rwanda	-9 to -5
Costa Rica	-5
North Macedonia	-6 to -4
Suriname	-8 to 0
Switzerland	3
Pakistan	9
Argentina	22
Canada	-18 to 25
Cambodia	38
Colombia	54
Peru	62
Tanzania	72
Ethiopia	-100 to 92
Indonesia	-15 to 214
Brazil	73 to 358
Nigeria	364

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Table S2: Regional forest-sector CO2 fluxes, assuming attainment of the regional aggregate NDC in 2030, relative to the 2020 flux estimate in GTM (negative values indicate sequestration).

	Upper Bound Target (MtCO2)	Lower Bound Target (MtCO2)
North Africa and the Middle East	-228	-36
Brazil	-437	-152
Canada	-3	40
Central America	-10	-10
China	-271	-271
East Asia	-13	6
Western Europe	-557	-556
Eastern Europe	-31	-29
Japan	-134	-134
Oceania	-162	-162
South America not including Brazil	396	403
Russia	-40	106
Southeast Asia	-899	-658
South Asia	-61	-39
Sub Saharan Africa	775	846
United States	-1078	-866